8.3 The One-Sample *t* Test

When n is small, the Central Limit Theorem (CLT) can no longer be invoked to justify the use of a large-sample test. We faced this same difficulty in obtaining a small-sample confidence interval (CI) for μ in Chapter 7. Our approach here will be the same one used there: We will assume that the population distribution is at least approximately normal and describe test procedures whose validity rests on this assumption. If an investigator has good reason to believe that the population distribution is quite nonnormal, a distribution-free test from Chapter 15 may be appropriate. Alternatively, a statistician can be consulted regarding procedures valid for specific families of population distributions other than the normal family. Or a bootstrap procedure can be developed.

The key result on which tests for a normal population mean are based was used in Chapter 7 to derive the one-sample t CI: If $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution, the standardized variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom (df). Consider testing H_0: $\mu = \mu_0$ using the test statistic $T = (\overline{X} - \mu_0)/(S/\sqrt{n})$. That is, the test statistic results from standardizing $\overline{X}$ under the assumption that H_0 is true (using $S/\sqrt{n}$, the estimated standard deviation of $\overline{X}$, rather than $\sigma/\sqrt{n}$). When H_0 is true, this test statistic has a t distribution with $n - 1$ df. Knowledge of the test statistic's distribution when H_0 is true (the "null distribution") allows us to determine the P-value.

The test statistic is really the same here as in the large-sample case but is labeled T to emphasize that the reference distribution for P-value determination is a t distribution with $n - 1$ df rather than the standard normal (z) distribution. Instead of being a z curve area as was the case for large-sample tests, the P-value will now be an area under the t_{n-1} curve (see Figure 8.6).

The One-Sample *t* Test

Null hypothesis: H_0: $\mu = \mu_0$

Test statistic value: $t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Alternative Hypothesis	P-Value Determination
H_a: $\mu > \mu_0$	Area under the t_{n-1} curve to the right of t
H_a: $\mu < \mu_0$	Area under the t_{n-1} curve to the left of t
H_a: $\mu \neq \mu_0$	$2 \cdot$ (Area under the t_{n-1} curve to the right of $\lvert t \rvert$)

Assumption: The data consists of a random sample from a normal population distribution.

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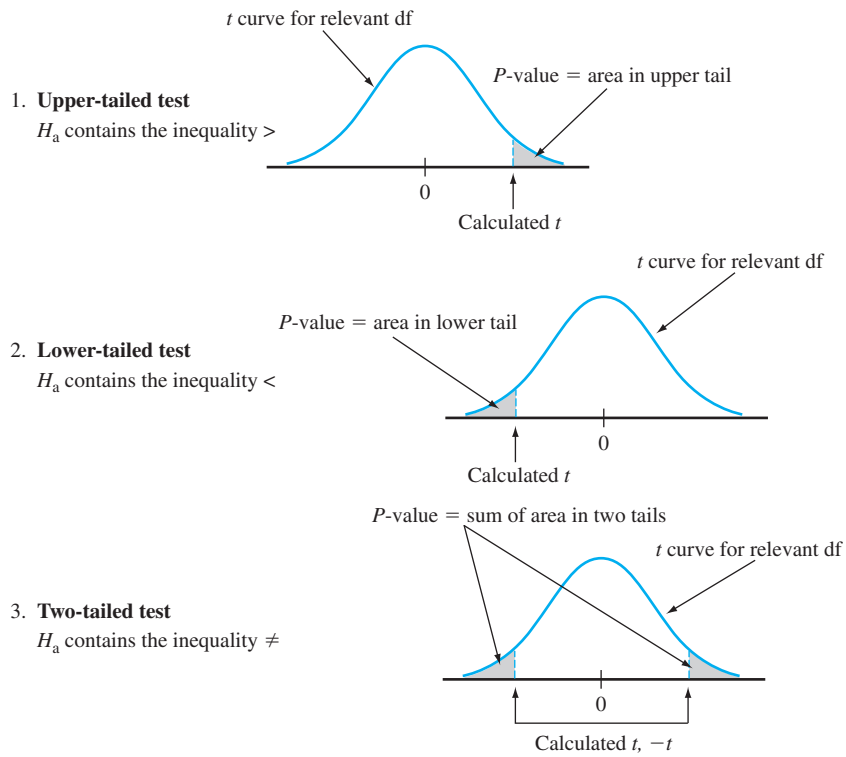

Figure 8.6 *P*-values for *t* tests

Unfortunately the table of t critical values that we used for confidence and prediction interval calculations in Chapter 7 does not provide much information about t curve tail areas. This is because for each t distribution there are values for only the seven most commonly used tail areas: .10, .05, .025, .01, .005, .001, and .0005. P-value determination would be straightforward if we had a table of tail areas (or alternatively, cumulative areas) that resembled our z table: for each different t distribution, the area under the corresponding curve to the right (or the left) of values 0.00, 0.01, 0.02, 0.03, … , 3.97, 3.98, 3.99, and finally 4.00. But this would necessitate an entire page of text for each different t distribution.

So we have included another t table in Appendix Table A.8. It contains a tabulation of upper-tail t curve areas but with less decimal accuracy than what the z table provides. Each different column of the table is for a different number of df, and the rows are for calculated values of the test statistic t ranging from 0.0 to 4.0 in increments of .1. For example, the number .074 appears at the intersection of the 1.6 row and the 8 df column. Thus the area under the 8 df curve to the right of 1.6 (an upper-tail area) is .074. Because t curves are symmetric about 0, .074 is also the area under the 8 df curve to the left of -1.6.

Suppose, for example, that a test of H_0: $\mu = 100$ versus H_a: $\mu > 100$ is based on the 8 df t distribution. If the calculated value of the test statistic is $t = 1.6$, then the P-value for this upper-tailed test is .074. Because .074 exceeds .05, we would not be able to reject H_0 at a significance level of .05. If the alternative hypothesis is H_a: $\mu < 100$ and a test based on 20 df yields $t = -3.2$, then Appendix Table A.7 shows that the P-value is the captured lower-tail area .002. The null hypothesis can be rejected at either level .05 or .01. In the next chapter, we will present a t test for hypotheses about a difference between two population means. Suppose the relevant hypotheses are H_0: $\mu_1 - \mu_2 = 0$ versus H_a: $\mu_1 - \mu_2 \neq 0$; the null hypothesis states that the means of the two

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populations are identical, whereas the alternative hypothesis states that they are different without specifying a direction of departure from H_0. If a t test is based on 20 df and $t = 3.2$, then the P-value for this two-tailed test is $2(.002) = .004$. This would also be the P-value for $t = -3.2$. The tail area is doubled because values both larger than 3.2 and smaller than -3.2 are more contradictory to H_0 than what was calculated (values farther out in *either* tail of the t curve).

EXAMPLE 8.9 Carbon nanofibers have potential application as heat-management materials, for composite reinforcement, and as components for nanoelectronics and photonics. The accompanying data on failure stress (MPa) of fiber specimens was read from a graph in the article **"Mechanical and Structural Characterization of Electrospun PAN-Derived Carbon Nanofibers" (*Carbon*, 2005: 2175–2185).**

300	312	327	368	400	425	470	556	573	575
580	589	626	637	690	715	757	891	900	

Summary quantities include $n = 19$, $\bar{x} = 562.68$, $s = 180.874$, $s/\sqrt{n} = 41.495$. Does the data provide compelling evidence for concluding that true average failure stress exceeds 500 MPa?

Figure 8.7 shows a normal probability plot of the data; the substantial linear pattern indicates that a normal population distribution of failure stress is quite plausible, giving us license to employ the one-sample t test (the box to the right of the plot gives information about a formal test of the hypothesis that the population distribution is normal; this will be discussed in Chapter 14).

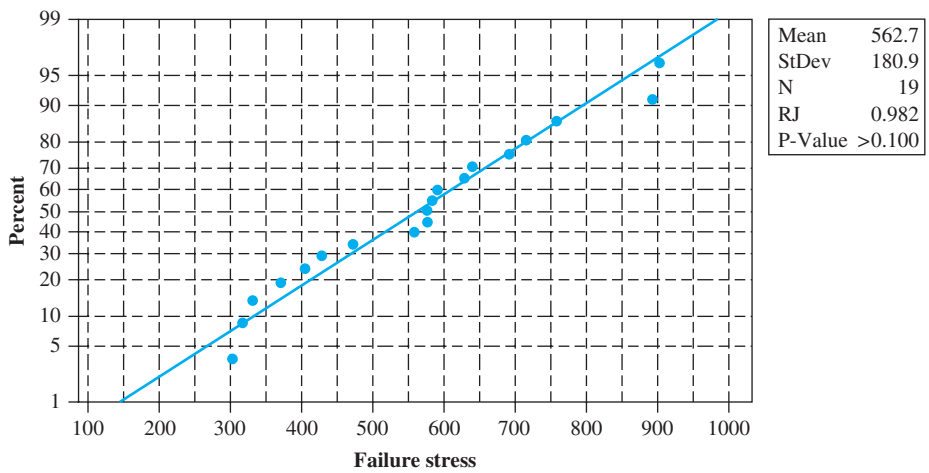

Figure 8.7 Normal probability plot of the failure stress data

Let's carry out a test of the relevant hypotheses using a significance level of .05.

1. The parameter of interest is μ = the true average failure stress
2. The null hypothesis is H_0: $\mu = 500$
3. The appropriate alternative hypothesis is H_a: $\mu > 500$ (so we'll believe that true average failure stress exceeds 500 only if the null hypothesis can be rejected).
4. The one-sample t test statistic is $T = (\bar{X} - 500)/(S/\sqrt{n})$. Its value t for the given data results from replacing $\bar{X}$ by $\bar{x}$ and S by s.
5. The test-statistic value is $t = (562.68 - 500)/41.495 = 1.51$

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6. The test is based on $19 - 1 = 18$ df. The entry in that column and the 1.5 row of Appendix Table A.8 is .075. Since the test is upper-tailed (because $>$ appears in H_a), it follows that P-value $\approx .075$ (Minitab says .074).
7. Because $.075 > .05$, there is not enough evidence to justify rejecting the null hypothesis at significance level .05. Rather than conclude that the true average failure stress exceeds 500, it appears that sampling variability provides a plausible explanation for the fact that the sample mean exceeds 500 by a rather substantial amount. ■

EXAMPLE 8.10 Many deleterious effects of smoking on health have been well documented. The article **"Smoking Abstinence Impairs Time Estimation Accuracy in Cigarette Smokers" (*Psychopharmacology Bull.*, 2003: 90–95)** described an investigation into whether time perception, an indicator of a person's ability to concentrate, is impaired during nicotine withdrawal. After a 24-hour smoking abstinence, each of 20 smokers was asked to estimate how much time had elapsed during a 45-second period. The following data on perceived elapsed time is consistent with summary quantities given in the cited article.

69	65	72	73	59	55	39	52	67	57
56	50	70	47	56	45	70	64	67	53

A normal probability plot of this data shows a very substantial linear pattern. Let's carry out a test of hypotheses at significance level .05 to decide whether true average perceived elapsed time differs from the known time 45.

1. μ = true average perceived elapsed time for all smokers exposed to the described experimental regimen
2. H_0: $\mu = 45$
3. H_0: $\mu \neq 45$
4. $t = (\bar{x} - 45)/(s/\sqrt{n})$
5. With $\bar{x} = 59.30$ and $s/\sqrt{n} = 9.84/\sqrt{20} = 2.200$, the test statistic value is $t = 14.3/2.200 = 6.50$.
6. The P-value for a two-tailed test is twice the area under the 19 df t curve to the right of 6.50. Since Table A.8 shows that the area under this t curve to the right of 4.0 is 0, the area to the right of 6.50 is certainly 0. The P-value is then $2(0) = 0$ (.00000 according to software).
7. A P-value as small as what we obtained argues very strongly for rejection of H_0 at any reasonable significance level, and in particular at significance level .05. The difference between the sample mean and its expected value when H_0 is true cannot plausibly be explained simply by chance variation. The true average perceived elapsed time is evidently something other than 45, so nicotine withdrawal does appear to impair perception of time. ■

β and Sample Size Determination

The calculation of β at the alternative value μ' for a normal population distribution with known σ was carried out by converting the inequality P-value $> \alpha$ to a statement about $\bar{x}$ (e.g., $\bar{x} < \mu_0 + z_\alpha \cdot \sigma/\sqrt{n}$) and then subtracting μ' to standardize correctly. An equivalent approach involves noting that when $\mu = \mu'$, the test statistic $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$ still has a normal distribution with variance 1, but now the mean value of Z is given by $(\mu' - \mu_0)/(\sigma/\sqrt{n})$. That is, when $\mu = \mu'$, the test

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statistic still has a normal distribution though not the standard normal distribution. Because of this, $\beta(\mu')$ is an area under the normal curve corresponding to mean value $(\mu' - \mu_0)/(\sigma/\sqrt{n})$ and variance 1. Both α and β involve working with normally distributed variables.

The calculation of $\beta(\mu')$ for the t test is much less straightforward. This is because the distribution of the test statistic $T = (\overline{X} - \mu_0)/(S/\sqrt{n})$ is quite complicated when H_0 is false and H_a is true. Thus, for an upper-tailed test, determining

$$\beta(\mu') = P(T < t_{\alpha, n-1} \text{ when } \mu = \mu' \text{ rather than } \mu_0)$$

involves integrating a very unpleasant density function. This must be done numerically. The results are summarized in graphs of β that appear in Appendix Table A.17. There are four sets of graphs, corresponding to one-tailed tests at level .05 and level .01 and two-tailed tests at the same levels.

To understand how these graphs are used, note first that both β and the necessary sample size n are as before functions not just of the absolute difference $|\mu_0 - \mu'|$ but of $d = |\mu_0 - \mu'|/\sigma$. Suppose, for example, that $|\mu_0 - \mu'| = 10$. This departure from H_0 will be much easier to detect (smaller β) when $\sigma = 2$, in which case μ_0 and μ' are 5 population standard deviations apart, than when $\sigma = 10$. The fact that β for the t test depends on d rather than just $|\mu_0 - \mu'|$ is unfortunate, since to use the graphs one must have some idea of the true value of σ. A conservative (large) guess for σ will yield a conservative (large) value of $\beta(\mu')$ and a conservative estimate of the sample size necessary for prescribed α and $\beta(\mu')$.

Once the alternative μ' and value of σ are selected, d is calculated and its value located on the horizontal axis of the relevant set of curves. The value of β is the height of the $n - 1$ df curve above the value of d (visual interpolation is necessary if $n - 1$ is not a value for which the corresponding curve appears), as illustrated in Figure 8.8.

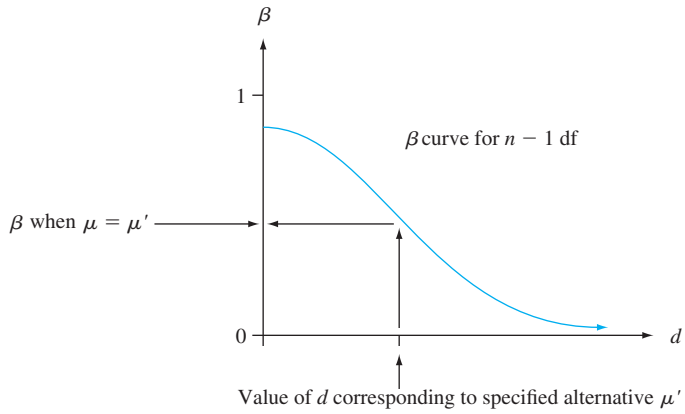

Figure 8.8 A typical β curve for the t test

Rather than fixing n (i.e., $n - 1$, and thus the particular curve from which β is read), one might prescribe both α (.05 or .01 here) and a value of β for the chosen μ' and σ. After computing d, the point (d, β) is located on the relevant set of graphs. The curve below and closest to this point gives $n - 1$ and thus n (again, interpolation is often necessary).

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EXAMPLE 8.11 The true average voltage drop from collector to emitter of insulated gate bipolar transistors of a certain type is supposed to be at most 2.5 volts. An investigator selects a sample of $n = 10$ such transistors and uses the resulting voltages as a basis for testing H_0: $\mu = 2.5$ versus H_a: $\mu > 2.5$ using a t test with significance level $\alpha = .05$. If the standard deviation of the voltage distribution is $\sigma = .100$, how likely is it that H_0 will not be rejected when in fact $\mu = 2.6$? With $d = |2.5 - 2.6|/.100 = 1.0$, the point on the β curve at 9 df for a one-tailed test with $\alpha = .05$ above 1.0 has a height of approximately .1, so $\beta \approx .1$. The investigator might think that this is too large a value of β for such a substantial departure from H_0 and may wish to have $\beta = .05$ for this alternative value of μ. Since $d = 1.0$, the point $(d, \beta) = (1.0, .05)$ must be located. This point is very close to the 14 df curve, so using $n = 15$ will give both $\alpha = .05$ and $\beta = .05$ when the value of μ is 2.6 and $\sigma = .10$. A larger value of σ would give a larger β for this alternative, and an alternative value of μ closer to 2.5 would also result in an increased value of β. ■

Most of the widely used statistical software packages are capable of calculating type II error probabilities. They generally work in terms of **power**, which is simply $1 - \beta$. A small value of β (close to 0) is equivalent to large power (near 1). A *powerful* test is one that has high power and therefore good ability to detect when the null hypothesis is false.

As an example, we asked Minitab to determine the power of the upper-tailed test in Example 8.11 for the three sample sizes 5, 10, and 15 when $\alpha = .05$, $\sigma = .10$, and the value of μ is actually 2.6 rather than the null value 2.5—a "difference" of. $2.6 - 2.5 = .1$. We also asked the software to determine the necessary sample size for a power of .9 ($\beta = .1$) and also .95. Here is the resulting output:

```
Power and Sample Size
Testing mean = null (versus > null)
Calculating power for mean = null + difference
Alpha = 0.05 Assumed standard deviation = 0.1

            Sample
Difference    Size     Power
0.1             5     0.579737
0.1            10     0.897517
0.1            15     0.978916

               Sample Target
                                        Actual
Difference      Size       Power        Power
0.1              11        0.90        0.924489
0.1              13        0.95        0.959703
```

The power for the sample size $n = 10$ is a bit smaller than .9. So if we insist that the power be at least .9, a sample size of 11 is required and the actual power for that n is roughly .92. The software says that for a target power of .95, a sample size of $n = 13$ is required, whereas eyeballing our β curves gave 15. When available, this type of software is more reliable than the curves. Finally, Minitab now also provides power curves for the specified sample sizes, as shown in Figure 8.9. Such curves illustrate how the power increases for each sample size as the actual value of μ moves farther and farther away from the null value.

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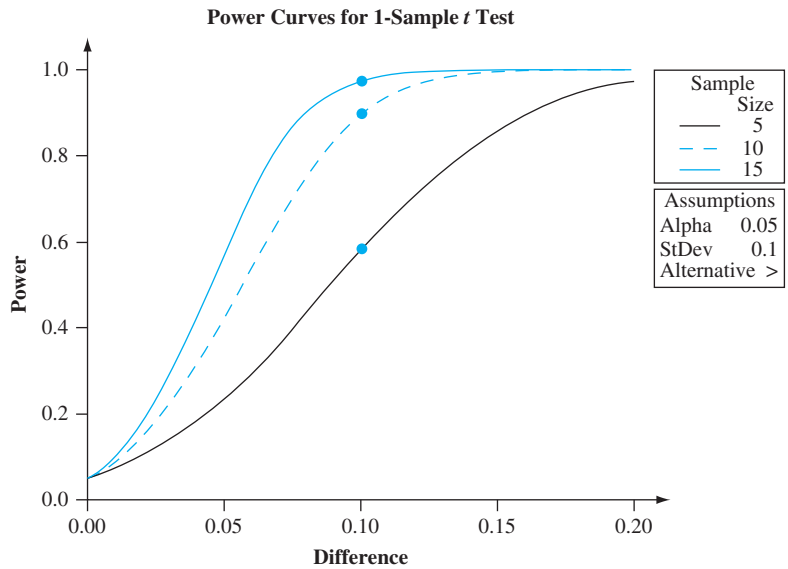

Figure 8.9 Power curves from Minitab for the t test of Example 8.11

Variation in *P*-values

The P-value resulting from carrying out a test on a selected sample is *not* the probability that H_0 is true, nor is it the probability of rejecting the null hypothesis. Once again, it is the probability, calculated assuming that H_0 is true, of obtaining a test statistic value at least as contradictory to the null hypothesis as the value that actually resulted. For example, consider testing H_0: $\mu = 50$ against H_0: $\mu < 50$ using a lower-tailed t test based on 20 df. If the calculated value of the test statistic is $t = -2.00$, then

$$\begin{aligned} P\text{-value} &= P(T < -2.00 \text{ when } \mu = 50) \\ &= \text{area under the } t_{20} \text{ to the left of } -2.00 = .030 \end{aligned}$$

But if a second sample is selected, the resulting value of t will almost surely be different from -2.00, so the corresponding P-value will also likely differ from .030. Because the test statistic value itself varies from one sample to another, the P-value will also vary from one sample to another. That is, the test statistic is a random variable, and so the P-value will also be a random variable. A first sample may give a P-value of .030, a second sample may result in a P-value of .117, a third may yield .061 as the P-value, and so on.

If H_0 is false, we hope the P-value will be close to 0 so that the null hypothesis can be rejected. On the other hand, when H_0 is true, we'd like the P-value to exceed the selected significance level so that the correct decision to not reject H_0 is made. The next example presents simulations to show how the P-value behaves both when the null hypothesis is true and when it is false.

EXAMPLE 8.12 The fuel efficiency (mpg) of any particular new vehicle under specified driving conditions may not be identical to the EPA figure that appears on the vehicle's sticker. Suppose that four different vehicles of a particular type are to be selected and driven over a certain course, after which the fuel efficiency of each one is to be determined.

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Let μ denote the true average fuel efficiency under these conditions. Consider testing H_0: $\mu = 20$ versus H_0: $\mu > 20$ using the one-sample t test based on the resulting sample. Since the test is based on $n - 1 = 3$ degrees of freedom, the P-value for an upper-tailed test is the area under the t curve with 3 df to the right of the calculated t.

Let's first suppose that the null hypothesis is true. We asked Minitab to generate 10,000 different samples, each containing 4 observations, from a normal population distribution with mean value $\mu = 20$ and standard deviation $\sigma = 2$. The first sample and resulting summary quantities were

$$x_1 = 20.830,\ x_2 = 22.232,\ x_3 = 20.276,\ x_4 = 17.718$$

$$\bar{x} = 20.264 \quad s = 1.8864 \quad t = \frac{20.264 - 20}{.1.8864/\sqrt{4}} = .2799$$

The P-value is the area under the 3-df t curve to the right of .2799, which according to Minitab is .3989. Using a significance level of .05, the null hypothesis would of course not be rejected. The values of t for the next four samples were -1.7591, .6082, $-.7020$, and 3.1053, with corresponding P-values .912, .293, .733, and .0265.

Figure 8.10(a) shows a histogram of the 10,000 P-values from this simulation experiment. About 4.5% of these P-values are in the first class interval from 0 to .05. Thus when using a significance level of .05, the null hypothesis is rejected in roughly 4.5% of these 10,000 tests. If we continued to generate samples and carry out the test for each sample at significance level .05, in the long run 5% of the P-values would be in the first class interval. This is because when H_0 is true and a test with significance level .05 is used, by definition the probability of rejecting H_0 is .05.

Looking at the histogram, it appears that the distribution of P-values is relatively flat. In fact, it can be shown that when H_0 is true, the probability distribution of the P-value is a uniform distribution on the interval from 0 to 1. That is, the density curve is completely flat on this interval, and thus must have a height of 1 if the total area under the curve is to be 1. Since the area under such a curve to the left of .05 is $(.05)(1) = .05$, we again have that the probability of rejecting H_0 when it is true that it is .05, the chosen significance level.

Now consider what happens when H_0 is false because $\mu = 21$. We again had Minitab generate 10,000 different samples of size 4 (each from a normal distribution with $\mu = 21$ and $\sigma = 2$), calculate $t = (\bar{x} - 20)/(s/\sqrt{4})$ for each one, and then determine the P-value. The first such sample resulted in $\bar{x} = 20.6411$, $s = .49637$, $t = 2.5832$, P-value $= .0408$. Figure 8.10(b) gives a histogram of the resulting P-values. The shape of this histogram is quite different from that of Figure 8.10(a)—there is a much greater tendency for the P-value to be small (closer to 0) when $\mu = 21$ than when $\mu = 20$. Again H_0 is rejected at significance level .05 whenever the P-value is at most .05 (in the first class interval). Unfortunately, this is the case for only about 19% of the P-values. So only about 19% of the 10,000 tests correctly reject the null hypothesis; for the other 81%, a type II error is committed. The difficulty is that the sample size is quite small and 21 is not very different from the value asserted by the null hypothesis.

Figure 8.10(c) illustrates what happens to the P-value when H_0 is false because $\mu = 22$ (still with $n = 4$ and $\sigma = 2$). The histogram is even more concentrated toward values close to 0 than was the case when $\mu = 21$. In general, as μ moves farther to the right of the null value 20, the distribution of the P-value will become more and more concentrated on values close to 0. Even here a bit fewer than 50% of the P-values are smaller than .05. So it is still slightly more likely than

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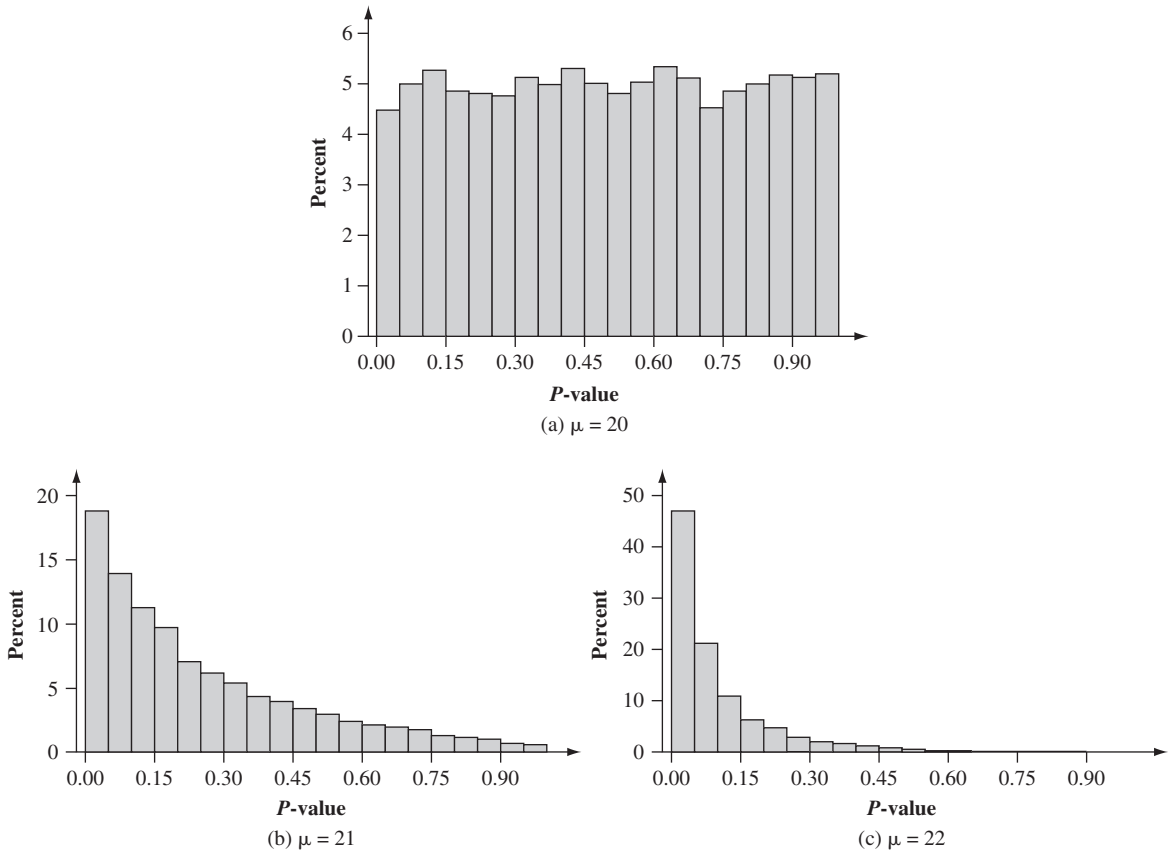

Figure 8.10 *P*-value simulation results for Example 8.12

not that the null hypothesis is incorrectly not rejected. Only for values of μ much larger than 20 (e.g., at least 24 or 25) is it highly likely that the P-value will be smaller than .05 and thus give the correct conclusion.

The big idea of this example is that because the value of any test statistic is random, the P-value will also be a random variable and thus have a distribution. The farther the actual value of the parameter is from the value specified by the null hypothesis, the more the distribution of the P-value will be concentrated on values close to 0 and the greater the chance that the test will correctly reject H_0 (corresponding to smaller β). ■

Whenever the observed value of a statistic such as $\overline{X}$ or $\hat{p}$ is reported, it is good statistical practice to include a quantitative measure of the statistic's precision, e.g., that the estimated standard error of $\overline{X}$ is $s/\sqrt{n}$. The P-value itself is a statistic—its value can be calculated once sample data is available and a particular test procedure is selected, and before such data is in hand, the P-value is subject to randomness. So it would be nice to have available σ_P or an estimate of this standard deviation.

Unfortunately the sampling distribution of a P-value is in general quite complicated. The simulation results of Example 8.12 suggest that the sampling distribution is quite skewed when H_0 is false (it is uniformly distributed on (0,1) when H_0 is true and the test statistic has a continuous distribution, e.g., a t distribution). A standard deviation is not as easy to interpret and use when there is substantial non-normality. The statisticians Dennis Boos and Leonard Stefanski investigated the random behavior of

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the P-value in their article **"*P*-Value Precision and Reproducibility" (*The American Statistician*, 2011: 213–221).** To address non-normality, they focused on the quantity –log(P-value). The log-transformed P-value does for many test procedures have approximately a normal distribution when n is large.

Suppose application of a particular test procedure to sample data results in a P-value of .001. Then H_0 would be rejected using either a significance level of .05 or .01. If a new sample from the same population distribution is then selected, how likely is it that the P-value for this new data will lead to rejection of H_0 at a significance level of .05 or .01? This is what the authors of the foregoing article meant by "reproducibility": How likely is it that a new sample will lead to the same conclusion as that reached using the original sample? The answer to this question depends on the population distribution, the sample size, and the test procedure used. Nevertheless, based on their investigations, the authors suggested the following general guidelines:

> If the P-value for the original data is .0001, then $P(\text{new } P\text{-value} \le .05) \approx .97$, whereas this probability is roughly .91 if the original P-value is .001 and it is roughly .73 when the original P-value is .01.

Particularly when the original P-value is around .01, there is a reasonably good chance that a new sample will not lead to rejection of H_0 at the 5% significance level. Thus unless the original P-value is really small, it would not be surprising to have a new sample contradict the inference drawn from the original data. A P-value not too much smaller than a chosen significance level such as .05 or .01 should be viewed with some caution!

EXERCISES Section 8.3 (29–41)

29. The true average diameter of ball bearings of a certain type is supposed to be .5 in. A one-sample t test will be carried out to see whether this is the case. What conclusion is appropriate in each of the following situations?

a. $n = 13, t = 1.6, \alpha = .05$
b. $n = 13, t = -1.6, \alpha = .05$
c. $n = 25, t = -2.6, \alpha = .01$
d. $n = 25, t = -3.9$

30. A sample of n sludge specimens is selected and the pH of each one is determined. The one-sample t test will then be used to see if there is compelling evidence for concluding that true average pH is less than 7.0. What conclusion is appropriate in each of the following situations?

a. $n = 6, t = -2.3, \alpha = .05$
b. $n = 15, t = -3.1, \alpha = .01$
c. $n = 12, t = -1.3, \alpha = .05$
d. $n = 6, t = .7, \alpha = .05$
e. $n = 6, \bar{x} = 6.68, s/\sqrt{n} = .0820$

31. The paint used to make lines on roads must reflect enough light to be clearly visible at night. Let μ denote the true average reflectometer reading for a new type of paint under consideration. A test of H_0: $\mu = 20$ versus H_a: $\mu > 20$ will be based on a random sample of size n from a normal population distribution. What conclusion is appropriate in each of the following situations?

a. $n = 15, t = 3.2, \alpha = .05$
b. $n = 9, t = 1.8, \alpha = .01$
c. $n = 24, t = -.2$

32. The relative conductivity of a semiconductor device is determined by the amount of impurity "doped" into the device during its manufacture. A silicon diode to be used for a specific purpose requires an average cut-on voltage of .60 V, and if this is not achieved, the amount of impurity must be adjusted. A sample of diodes was selected and the cut-on voltage was determined. The accompanying SAS output resulted from a request to test the appropriate hypotheses.

N	Mean	Std Dev	T	Prob. > |T|
15	0.0453333	0.0899100	1.9527887	0.0711

[*Note:* SAS explicitly tests H_0: $\mu = 0$, so to test H_0: $\mu = .60$, the null value .60 must be subtracted from each x_i; the reported mean is then the average of the $(x_i - .60)$ values. Also, SAS's P-value is always for a two-tailed test.] What would be concluded for a significance level of .01? .05? .10?

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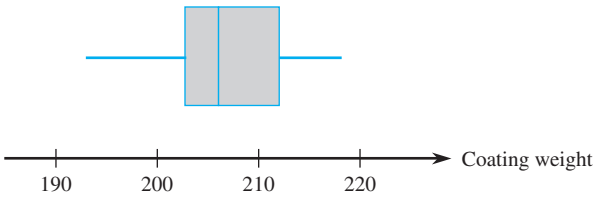
33. The article **"The Foreman's View of Quality Control" (*Quality Engr.*, 1990: 257–280)** described an investigation into the coating weights for large pipes resulting from a galvanized coating process. Production standards call for a true average weight of 200 lb per pipe. The accompanying descriptive summary and boxplot are from Minitab.

Variable	N	Mean	Median	TrMean	StDev	SEMean
ctg wt	30	206.73	206.00	206.81	6.35	1.16

Variable	Min	Max	Q1	Q3
ctg wt	193.00	218.00	202.75	212.00



a. What does the boxplot suggest about the status of the specification for true average coating weight?

b. A normal probability plot of the data was quite straight. Use the descriptive output to test the appropriate hypotheses.

34. The following observations are on stopping distance (ft) of a particular truck at 20 mph under specified experimental conditions **("Experimental Measurement of the Stopping Performance of a Tractor-Semitrailer from Multiple Speeds," NHTSA, DOT HS 811 488, June 2011):**

32.1 30.6 31.4 30.4 31.0 31.9

The cited report states that under these conditions, the maximum allowable stopping distance is 30. A normal probability plot validates the assumption that stopping distance is normally distributed.

a. Does the data suggest that true average stopping distance exceeds this maximum value? Test the appropriate hypotheses using $\alpha = .01$.

b. Determine the probability of a type II error when $\alpha = .01$, $\sigma = .65$, and the actual value of μ is 31. Repeat this for $\mu = 32$ (use either statistical software or Table A.17).

c. Repeat (b) using $\sigma = .80$ and compare to the results of (b).

d. What sample size would be necessary to have $\alpha = .01$ and $\beta = .10$ when $\mu = 31$ and $\sigma = .65$?

35. The article **"Uncertainty Estimation in Railway Track Life-Cycle Cost" (*J. of Rail and Rapid Transit*, 2009)** presented the following data on time to repair (min) a rail break in the high rail on a curved track of a certain railway line.

159 120 480 149 270 547 340 43 228 202 240 218

A normal probability plot of the data shows a reasonably linear pattern, so it is plausible that the population distribution of repair time is at least approximately normal. The sample mean and standard deviation are 249.7 and 145.1, respectively.

a. Is there compelling evidence for concluding that true average repair time exceeds 200 min? Carry out a test of hypotheses using a significance level of .05.

b. Using $\sigma = 150$, what is the type II error probability of the test used in (a) when true average repair time is actually 300 min? That is, what is $\beta(300)$?

36. Have you ever been frustrated because you could not get a container of some sort to release the last bit of its contents? The article **"Shake, Rattle, and Squeeze: How Much Is Left in That Container?" (*Consumer Reports*, May 2009: 8)** reported on an investigation of this issue for various consumer products. Suppose five 6.0 oz tubes of toothpaste of a particular brand are randomly selected and squeezed until no more toothpaste will come out. Then each tube is cut open and the amount remaining is weighed, resulting in the following data (consistent with what the cited article reported): .53, .65, .46, .50, .37. Does it appear that the true average amount left is less than 10% of the advertised net contents?

a. Check the validity of any assumptions necessary for testing the appropriate hypotheses.

b. Carry out a test of the appropriate hypotheses using a significance level of .05. Would your conclusion change if a significance level of .01 had been used?

c. Describe in context type I and II errors, and say which error might have been made in reaching a conclusion.

37. The accompanying data on cube compressive strength (MPa) of concrete specimens appeared in the article **"Experimental Study of Recycled Rubber-Filled High-Strength Concrete" (*Magazine of Concrete Res.*, 2009: 549–556):**

112.3	97.0	92.7	86.0	102.0
99.2	95.8	103.5	89.0	86.7

a. Is it plausible that the compressive strength for this type of concrete is normally distributed?

b. Suppose the concrete will be used for a particular application unless there is strong evidence that true average strength is less than 100 MPa. Should the concrete be used? Carry out a test of appropriate hypotheses.

38. A random sample of soil specimens was obtained, and the amount of organic matter (%) in the soil was determined for each specimen, resulting in the accompanying data (from **"Engineering Properties of Soil," *Soil Science*, 1998: 93–102).**

1.10	5.09	0.97	1.59	4.60	0.32	0.55	1.45
0.14	4.47	1.20	3.50	5.02	4.67	5.22	2.69
3.98	3.17	3.03	2.21	0.69	4.47	3.31	1.17
0.76	1.17	1.57	2.62	1.66	2.05		

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The values of the sample mean, sample standard deviation, and (estimated) standard error of the mean are 2.481, 1.616, and .295, respectively. Does this data suggest that the true average percentage of organic matter in such soil is something other than 3%? Carry out a test of the appropriate hypotheses at significance level .10. Would your conclusion be different if $\alpha = .05$ had been used? [*Note:* A normal probability plot of the data shows an acceptable pattern in light of the reasonably large sample size.]

39. Reconsider the accompanying sample data on expense ratio (%) for large-cap growth mutual funds first introduced in Exercise 1.53.

0.52	1.06	1.26	2.17	1.55	0.99	1.10	1.07	1.81	2.05
0.91	0.79	1.39	0.62	1.52	1.02	1.10	1.78	1.01	1.15

A normal probability plot shows a reasonably linear pattern.

a. Is there compelling evidence for concluding that the population mean expense ratio exceeds 1%? Carry out a test of the relevant hypotheses using a significance level of .01.

b. Referring back to (a), describe in context type I and II errors and say which error you might have made in reaching your conclusion. The source from which the data was obtained reported that $\mu = 1.33$ for the population of all 762 such funds. So did you actually commit an error in reaching your conclusion?

c. Supposing that $\sigma = .5$, determine and interpret the power of the test in (a) for the actual value of μ stated in (b).

40. Polymer composite materials have gained popularity because they have high strength to weight ratios and are relatively easy and inexpensive to manufacture. However, their nondegradable nature has prompted development of environmentally friendly composites using natural materials. The article **"Properties of Waste Silk Short Fiber/Cellulose Green Composite Films" (*J. of Composite Materials*, 2012: 123–127)** reported that for a sample of 10 specimens with 2% fiber content, the sample mean tensile strength (MPa) was 51.3 and the sample standard deviation was 1.2. Suppose the true average strength for 0% fibers (pure cellulose) is known to be 48 MPa. Does the data provide compelling evidence for concluding that true average strength for the WSF/cellulose composite exceeds this value?

41. A spectrophotometer used for measuring CO concentration [ppm (parts per million) by volume] is checked for accuracy by taking readings on a manufactured gas (called span gas) in which the CO concentration is very precisely controlled at 70 ppm. If the readings suggest that the spectrophotometer is not working properly, it will have to be recalibrated. Assume that if it is properly calibrated, measured concentration for span gas samples is normally distributed. On the basis of the six readings—85, 77, 82, 68, 72, and 69—is recalibration necessary? Carry out a test of the relevant hypotheses using $\alpha = .05$.

8.4 Tests Concerning a Population Proportion

Let p denote the proportion of individuals or objects in a population who possess a specified property (e.g., college students who graduate without any debt, or computers that do not need service during the warranty period). If an individual or object with the property is labeled a success (S), then p is the population proportion of successes. Tests concerning p will be based on a random sample of size n from the population. Provided that n is small relative to the population size, X (the number of S's in the sample) has (approximately) a binomial distribution. Furthermore, if n itself is large [$np \geq 10$ and $n(1-p) \geq 10$], both X and the estimator $\hat{p} = X/n$ are approximately normally distributed. We first consider large-sample tests based on this latter fact and then turn to the small-sample case that directly uses the binomial distribution.

Large-Sample Tests

Large-sample tests concerning p are a special case of the more general large-sample procedures for a parameter θ. Let $\hat{\theta}$ be an estimator of θ that is (at least approximately) unbiased and has approximately a normal distribution. The null hypothesis has the form H_0: $\theta = \theta_0$ where θ_0 denotes a number

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